NATIONAL ECONOMICS UNIVERSITY
COLLEGE OF TECHNOLOGY

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MID-TERM ASSIGNMENT

COURSE: Data Analysis with Python

TOPIC: CLUSTERING EMPLOYEES BY WORK BEHAVIOUR AND SATISFACTION – HR ANALYTICS CASE STUDY

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Abstract

This report applies unsupervised learning to the *HR Analytics Case Study* dataset, which describes about 4,410 employees of a company through demographic, job, survey and attendance (swipe-in/swipe-out) records.

The goal is to discover naturally occurring employee segments without using any pre-defined label. We integrate the relational files, clean and engineer features (including average daily working hours derived from attendance logs), and compare three clustering algorithms — K-Means, Hierarchical (Ward) clustering and DBSCAN — using the Silhouette score, the Elbow method, Davies–Bouldin and Calinski–Harabasz indices.

The resulting segments are profiled and cross-checked against the `Attrition` column (used only for external validation) to derive practical HR recommendations.

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1 Initial Data Inspection

Before data preprocessing, an initial inspection was conducted to understand the structure and quality of all five datasets. The inspection focused on the consistency of the files, the relationships between them, file encoding, data types, missing values, and duplicate records.

1.1 Dataset Overview

The HR Analytics dataset consists of five relational files: `general`, `employee`, `manager`, `in_time`, and `out_time`. Each file contains 4,410 records.

Table 1 summarizes the basic information of the five datasets.

Table 1: Overview of the five datasets

Criteria	General	Employee	Manager	In-time	Out-time
Records	4,410	4,410	4,410	4,410	4,410
Attributes	24	4	3	262	262
Missing values	28	83	0	109080	109080
Duplicate records	0	0	0	0	0

Across the five files, there are 555 attributes in total. However, the 22,050 records represent the sum of records across files, not the number of unique employees, since the same employees are represented in multiple relational files.

1.2 Column Consistency

The five files describe different information, so they cannot be concatenated; they must be merged later. Consistency was therefore checked in two ways. First, the numbers of rows and columns were compared (Table 1): every file has 4,410 rows, while the number of columns differs by file (24, 4, 3, 262 and 262). Second, the column names of `in_time` and `out_time`, which have the same structure (one column per calendar day), were compared. They contain the same 262 column names in the same order, so the two files can be processed in parallel when computing working hours. No column name other than `EmployeeID` is shared between the other files, so no naming conflict (e.g. `_x`, `_y` suffixes) will appear after merging.

1.3 Relationships Between Files

The common column of the five files is `EmployeeID`. In `in_time` and `out_time` this column is stored in an unnamed first column (`Unnamed: 0`) and is renamed to `EmployeeID`. Table 2 shows the key checks.

Table 2: Primary key / foreign key check on `EmployeeID`

File	Null keys	Duplicate keys	IDs not in General	General IDs missing
General	0	0	0	0
Employee	0	0	0	0
Manager	0	0	0	0
In-time	0	0	0	0
Out-time	0	0	0	0

`EmployeeID` is the **primary key** of the `general` dataset (unique and non-null) and the **foreign key** in the other four files, so the relationship is one-to-one. A test merge of `general`, `employee` and `manager` on `EmployeeID` returned 4,410 rows, i.e. no record was lost or duplicated.

1.4 File Encoding

The encoding of each CSV file was tested by decoding the raw bytes. All five files were successfully decoded as UTF-8 (no BOM, no decoding error), so no special encoding argument is needed when calling `pd.read_csv()` and no font or character errors are expected.

1.5 Data Types

The data types of each dataset were examined using the `info()` method.

Table 3: Distribution of data types across datasets

Dataset	int64	float64	str	Total attributes
General	14	2	8	24
Employee	1	3	0	4
Manager	3	0	0	3
In-time	1	12	249	262
Out-time	1	12	249	262
Total	**20**	**29**	**506**	**555**

The `general` dataset contains demographic and job-related information. The `employee` dataset provides employee satisfaction and work-life balance information, while the `manager` dataset contains job involvement and performance ratings.

The `in_time` and `out_time` datasets each contain 262 attributes, including 249 string attributes, 12 floating-point attributes, and 1 integer attribute. These two files contain attendance-related records and will be examined further during feature engineering.

1.6 Missing Values

Missing values were identified in the `general` and `employee` datasets.

1.6.1 General Dataset

The `general` dataset contains 28 missing values across two attributes.

Table 4: Missing values in the General dataset

Attribute	Missing values	Percentage
NumCompaniesWorked	19	0.43%
TotalWorkingYears	9	0.20%
Total	**28**	**0.63%**

1.6.2 Employee Dataset

The `employee` dataset contains 83 missing values across three attributes.

Table 5: Missing values in the Employee dataset

Attribute	Missing values	Percentage
EnvironmentSatisfaction	25	0.57%
JobSatisfaction	20	0.45%
WorkLifeBalance	38	0.86%
Total	**83**	**1.88%**

1.6.3 Manager Dataset

The `manager` dataset contains three attributes: `EmployeeID`, `JobInvolvement`, and `PerformanceRating`. All 4,410 records are non-null, indicating that no missing values were detected.

1.6.4 Attendance Datasets

The `in_time` and `out_time` datasets each contain 262 attributes and 4,410 records. The available inspection output does not provide the non-null counts for individual columns. Therefore, their exact numbers of missing values need to be calculated separately before preprocessing.

1.7 Duplicate Records

Duplicate records were checked in all five datasets. The results showed that no duplicate records were detected in any of the files.

1.8 Initial Data Quality Assessment

Based on the initial inspection, the following observations were identified:

- All five datasets contain 4,410 records each, with a total of 555 attributes.
- The datasets contain numerical and string data types, requiring appropriate preprocessing techniques.
- The `general` dataset contains 28 missing values, while the `employee` dataset contains 83 missing values.
- The `manager` dataset contains no missing values.
- The exact number of missing values in the attendance datasets still needs to be determined.
- No duplicate records were detected in the five datasets.
- `EmployeeID` links all files (primary key in `general`, foreign key elsewhere), and the files can be read with UTF-8 encoding.

These findings provide an overview of the data structure and initial data quality. The next step is to examine the attendance records, handle missing values, and prepare suitable features for clustering.

2 Data Cleaning

The inspection in the previous section showed missing values in the `general`, `employee`, `in_time` and `out_time` files, no duplicate records, and attendance dates stored as strings. This section describes how each problem was handled to obtain the final table `df_clean`. The pipeline is implemented in the accompanying notebook.

2.1 Merging the Tables

The `general`, `employee` and `manager` files were joined on `EmployeeID` with `validate="one_to_one"`, which raises an error if any key is repeated. The merged table keeps all 4,410 employees and contains 28 columns. The two attendance files are processed separately (Section 2.4) and joined at the end.

2.2 Removing Duplicates

Duplicates were checked again after merging, because joins can silently multiply rows. Two criteria were used: fully identical rows and repeated `EmployeeID`. Both checks returned 0, so no record was removed and the table keeps 4,410 employees.

2.3 Handling Missing Values

Missing values were not treated in the same way everywhere. The decision depends on *why* a value is missing.

Table 6: Missing-value handling strategy

Dataset	Attribute	Missing	Strategy
General	NumCompaniesWorked	19	Median imputation
General	TotalWorkingYears	9	Median imputation
Employee	EnvironmentSatisfaction	25	Mode imputation
Employee	JobSatisfaction	20	Mode imputation
Employee	WorkLifeBalance	38	Mode imputation
Manager	—	0	No action
In/Out-time	all date columns	109,080 each	Not imputed: interpreted as absence and converted to features

2.3.1 General and Employee datasets

The General and Employee datasets contain 28 (0.63%) and 83 (1.88%) missing values, respectively. Because these counts are small, all rows were retained instead of being deleted, so that the other attributes of the affected employees are not lost. The missing values were imputed with a statistic chosen according to the nature of each variable.

Count variables: median. `NumCompaniesWorked` and `TotalWorkingYears` are non-negative counts. Figure 1 shows that both are right-skewed (skewness 1.12 and 1.03): most employees have low values, while a small number have very high ones (up to 40 years of experience and up to 9 previous companies) that form a long upper tail. These extreme values pull the mean above the median (11.28 vs. 10 years; 2.69 vs. 2 companies), so the mean no longer describes a typical employee, and for an integer count it also produces non-integer values. The median is not affected by the tail, so it was used.

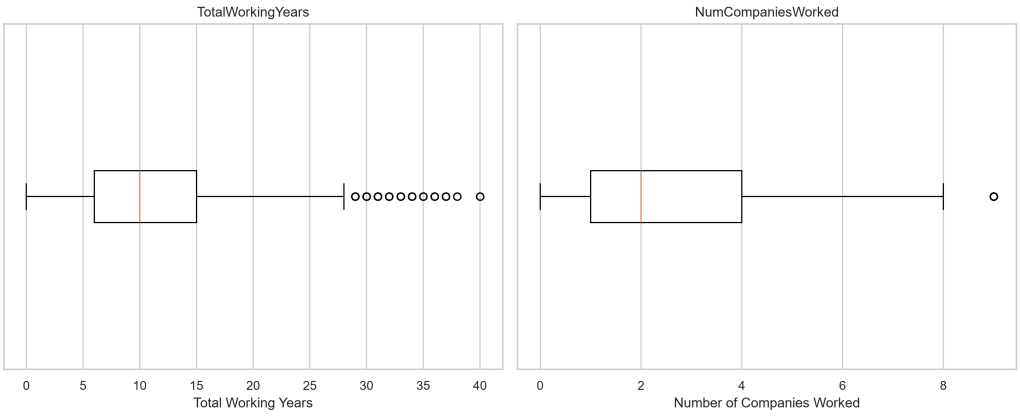


Figure 1: Distribution of `TotalWorkingYears` and `NumCompaniesWorked`, with mean (dashed) and median (solid).

Ordinal survey scores: mode. `EnvironmentSatisfaction`, `JobSatisfaction` and `WorkLifeBalance` are ordinal scores from 1 to 4. Figure 2 shows that their means (2.72, 2.73 and 2.76) fall between two levels and do not correspond to any valid answer. The mode (3, 4 and 3) is the level chosen most often, is always a valid score, and keeps the discrete nature of the variable, so it was used instead of the mean.

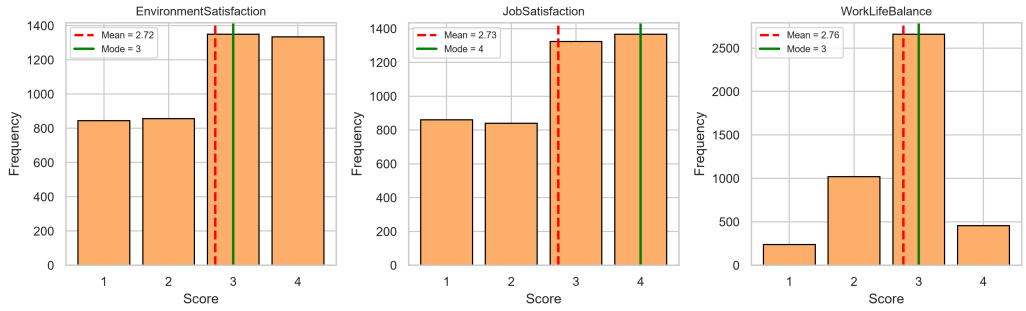


Figure 2: Distribution of the three survey scores, with mean (dashed) and mode (solid).

2.4 Processing Attendance Data

Each of `in_time` and `out_time` has 261 date columns and 109,080 missing cells. These are *not* random gaps. A cell is empty when the employee did not swipe on that day, that is, on a public holiday, on leave or when absent. Filling them with a mean or median would invent working hours that never happened. The cells were therefore handled in five steps:

Step 1. Convert to datetime. All 261 date columns were parsed with `pd.to_datetime`.

Step 2. Remove non-working days. Days on which no employee has a swipe-in and swipe-out are public holidays or closed days and carry no information. 12 such columns (the float columns in Table 3) were dropped. They account for $12 \times 4{,}410 = 52{,}920$ of the missing cells in each file, and the remaining $109{,}080 - 52{,}920 = 56{,}160$ cells are individual absences. The number of working days is therefore $N_{\text{work}} = 249$.

Step 3. Validate records. A day is valid only if both swipe-in and swipe-out exist and the duration lies in $(0, 24]$ hours. The check found 0 days with only one of the two swipes and 0 days with an invalid duration, so no record had to be discarded.

Step 4. Aggregate per employee. The remaining missing days are skipped (not filled) when computing the features below.

Step 5. Join. The three features are merged into the main table on `EmployeeID` (one-to-one).

$$h_{i,d} = t_{i,d}^{\text{out}} - t_{i,d}^{\text{in}} \quad \text{(hours, valid days only)}, \tag{1}$$

$$\text{AvgWorkHours}_i = \frac{1}{|D_i|} \sum_{d \in D_i} h_{i,d}, \tag{2}$$

$$\text{AbsentDays}_i = N_{\text{work}} - |D_i|, \qquad \text{AbsentRate}_i = \frac{\text{AbsentDays}_i}{N_{\text{work}}}, \tag{3}$$

where D_i is the set of valid days of employee i. Every employee has at least one valid day, so no median fallback was needed for these features.

Table 7: Summary of the engineered attendance features

Feature	**Mean**	**Std**	**Min**	**Median**	**Max**
AvgWorkHours	7.70	1.34	5.95	7.41	11.03
AbsentDays	12.74	5.50	1	13	24
AbsentRate	0.051	0.022	0.004	0.052	0.096

After this step the dataset contains **no missing values** (Table 9).

2.5 Standardizing Data Types

Table 8: Data type conversions

Attributes	**Before**	**After**	**Reason**
261 date columns of in/out-time	string	datetime	Needed to compute working hours
Survey scores, `NumCompaniesWorked`, `TotalWorkingYears`	float64	int64	Imputed values are whole numbers
`BusinessTravel`, `Department`, `EducationField`, `Gender`, `JobRole`, `MaritalStatus`	string	category	Memory and correct encoding
`Attrition`	string (Yes/No)	int64 (1/0)	External validation only

Three constant attributes (`EmployeeCount`, `Over18`, `StandardHours`) have a single value for all employees. They cannot separate clusters and were dropped.

A logical consistency check compared `YearsAtCompany` with `TotalWorkingYears`. Two employees (0.05%) have more years at the company than total working years. This is negligible and these records were kept unchanged.

2.6 Summary of Data Cleaning

Table 9: Dataset before and after cleaning

Criteria	Before	After
Employees (rows)	4,410	4,410
Missing values (general + survey)	111	0
Missing values (attendance, per file)	109,080	0 (converted to features)
Duplicate records	0	0
Constant attributes	3	0
Attributes (final table)	555 (5 files)	29

The 29 attributes of `df_clean` are: 21 attributes of `general` (24 minus 3 constant ones), 3 survey scores from `employee`, 2 ratings (`JobInvolvement`, `PerformanceRating`) from `manager`, and 3 attendance features. The cleaned table has one row per employee and is ready for exploratory data analysis and feature selection.

Listing 1: Core of the attendance cleaning step

```
in_dt  = in_df.apply(pd.to_datetime, errors="coerce")
out_dt = out_df.apply(pd.to_datetime, errors="coerce")

# drop public holidays: days with no swipe for every employee
holiday = in_dt.columns[in_dt.isna().all() & out_dt.isna().all()]
in_dt, out_dt = in_dt.drop(columns=holiday), out_dt.drop(columns=holiday
    )

# keep only days with both in and out
half = in_dt.isna() ^ out_dt.isna()
in_dt, out_dt = in_dt.mask(half), out_dt.mask(half)

hours = (out_dt - in_dt).apply(lambda s: s.dt.total_seconds() / 3600)
hours = hours.mask((hours <= 0) | (hours > 24))

att = pd.DataFrame({
    "AvgWorkHours": hours.mean(axis=1),          # NaN days skipped
    "AbsentDays":   hours.isna().sum(axis=1),
})
att["AbsentRate"] = att["AbsentDays"] / hours.shape[1]
```

3 Exploratory Data Analysis and Feature Selection

This section examines the cleaned table `df_clean`, selects the variables used for clustering, and prepares them (encoding, scaling, dimensionality reduction). No target variable exists in clustering, so the EDA looks for the distributions and relationships that decide which

fields can separate groups of employees. `Attrition` is not used here; it is kept for the external validation of the clusters.

3.1 Distribution Analysis

3.1.1 Age, income, tenure, distance and working hours

Table 10 and Figure 3 describe the main numeric variables.

Table 10: Summary statistics of the main numeric variables

Variable	Mean	Std	Min	Median	Max	Skew
Age	36.92	9.13	18	36	60	0.41
MonthlyIncome	65,029	47,069	10,090	49,190	199,990	1.37
TotalWorkingYears	11.28	7.77	0	10	40	1.12
YearsAtCompany	7.01	6.13	0	5	40	1.76
DistanceFromHome	9.19	8.11	1	7	29	0.96
AvgWorkHours	7.70	1.34	5.95	7.41	11.03	0.86

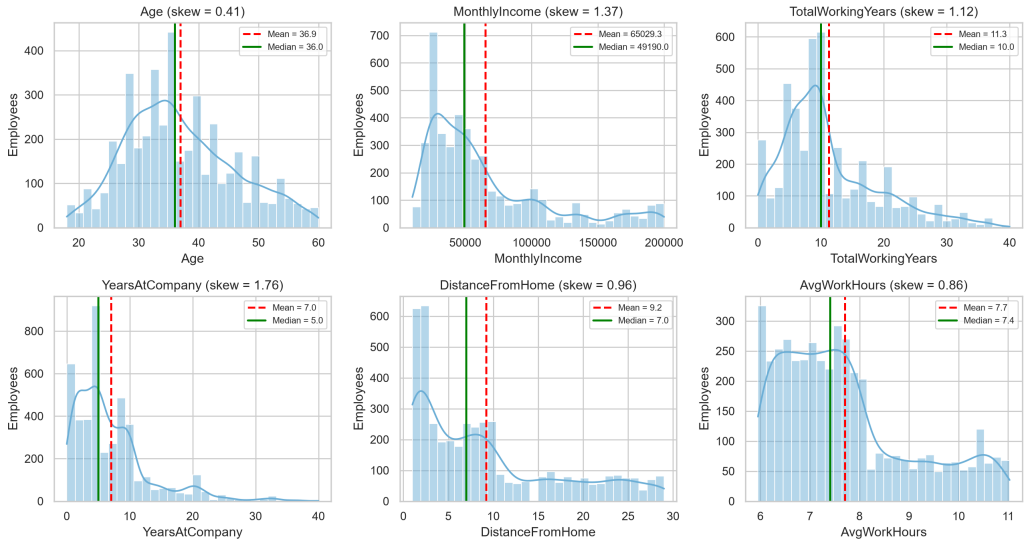


Figure 3: Distribution of the main numeric variables with mean (dashed) and median (solid).

- **Age** is almost symmetric (skew 0.41) and ranges from 18 to 60, so no transformation is needed.
- **MonthlyIncome** is right-skewed (skew 1.37): the mean (65,029) is well above the median (49,190), and a few employees earn close to 200,000.
- **TotalWorkingYears** and **YearsAtCompany** are also right-skewed (1.12 and 1.76). Half of the employees have been at the company for 5 years or less, while a small group reaches 40 years.

- **DistanceFromHome** is slightly right-skewed (0.96); half of the employees live within 7 km, the maximum is 29 km.
- **AvgWorkHours** (from the attendance logs) has a mean of 7.70 hours and a range of 5.95–11.03 hours. About 18% of employees (813) average more than 9 hours per day, which is the overload signal sought for clustering.

The box plots in Figure 4 show the same upper tails as outlier points. These values are real (senior staff, very long careers) and are exactly the groups clustering should detect, so they are *not removed*; the skew is reduced by a log transformation instead (Section 3.4).

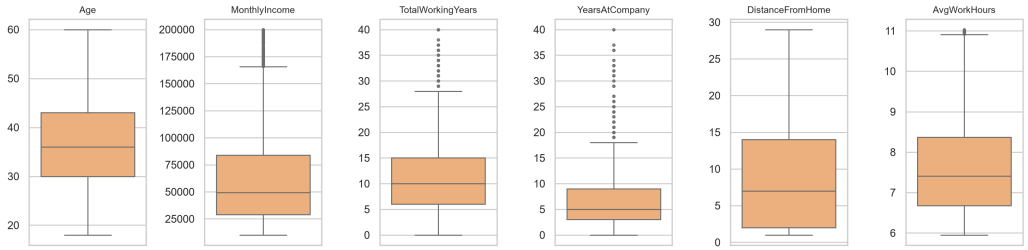


Figure 4: Box plots of the main numeric variables.

3.1.2 Satisfaction and engagement surveys

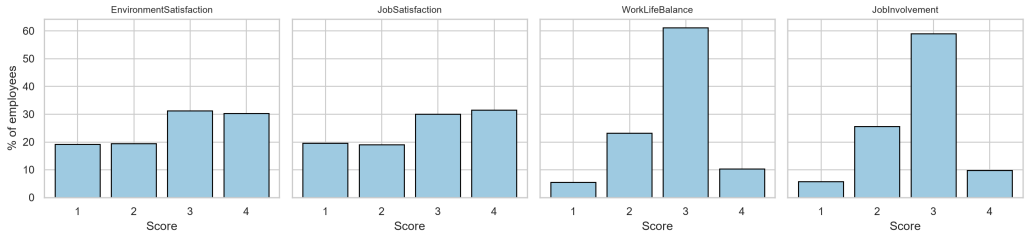


Figure 5: Share of employees at each score (1–4) of the four survey variables.

Figure 5 shows two different patterns. `EnvironmentSatisfaction` and `JobSatisfaction` are spread over all four levels: about 39% of employees give a low score (1–2) and about 61% a high score (3–4), so a clearly dissatisfied group exists. `WorkLifeBalance` and `JobInvolvement` are concentrated at level 3 (61.2% and 59.0%), and only 5.4% and 5.6% of employees choose level 1. These two variables separate groups less strongly, but their extreme levels point to small and potentially at-risk segments.

3.2 Correlation Analysis

Spearman correlation was used because many variables are ordinal or skewed. Figure 6 shows the matrix and Table 11 lists the pairs with $|\rho| \geq 0.5$.

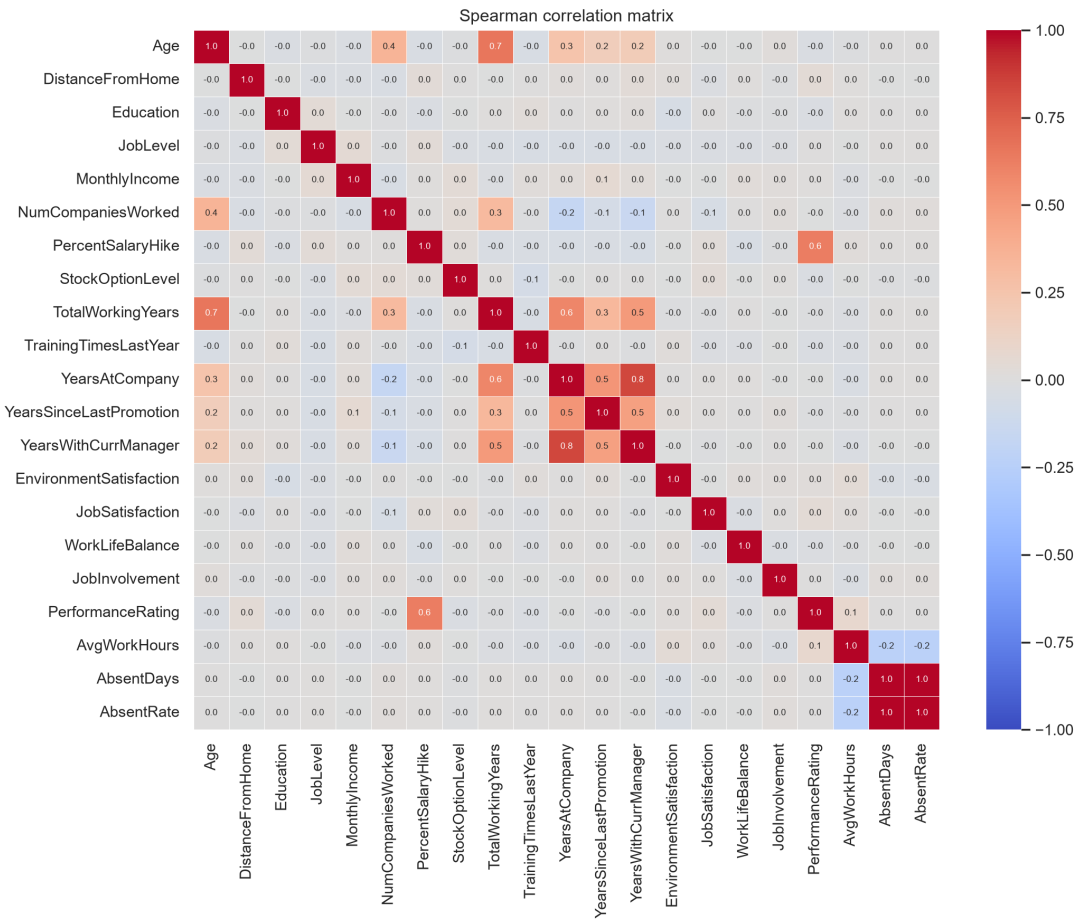


Figure 6: Spearman correlation matrix of the numeric variables.

Table 11: Strongly correlated pairs ($|\rho| \geq 0.5$)

Variable 1	**Variable 2**	**Spearman ρ**
AbsentDays	AbsentRate	1.00
YearsAtCompany	YearsWithCurrManager	0.84
Age	TotalWorkingYears	0.66
PercentSalaryHike	PerformanceRating	0.63
TotalWorkingYears	YearsAtCompany	0.59
YearsAtCompany	YearsSinceLastPromotion	0.52

Three observations drive the feature selection:

1. The strongly correlated pairs are redundant measurements of the same idea (tenure, age, absence). Keeping both variables would count that idea twice in a distance-based model.

2. Outside these pairs the variables are almost uncorrelated. For example `MonthlyIncome` is unrelated to `JobLevel` ($\rho = 0.05$) and to tenure, and `AvgWorkHours` is unrelated to the satisfaction scores ($|\rho| \leq 0.05$). Income therefore acts as an independent dimension, and a single "senior and well paid" axis should not be expected.

3. The only noticeable relation of `AvgWorkHours` is with `AbsentRate` ($\rho = -0.23$): employees who work longer days are absent less often.

3.3 Feature Selection

The features were chosen to describe the three kinds of segments targeted by the project: senior and well-paid staff, newcomers and employees at burnout risk, and employees with heavy workload but low satisfaction. Table 12 lists the 13 selected features by group.

Table 12: Selected clustering features

Group	**Features**	**Purpose**
Seniority & pay	MonthlyIncome, JobLevel, TotalWorkingYears, YearsAtCompany, YearsSinceLastPromotion	Senior vs. new staff, stagnation
Workload & attendance	AvgWorkHours, AbsentRate	Overload and absence
Satisfaction & engagement	EnvironmentSatisfaction, JobSatisfaction, WorkLifeBalance, JobInvolvement	Well-being, engagement
Commute & travel	DistanceFromHome, BusinessTravel	Extra load outside working hours

Table 13: Variables not used as clustering input

Variable	**Reason**
EmployeeID, Attrition	Identifier; Attrition is kept for external validation
AbsentDays	Duplicate of AbsentRate ($\rho = 1.00$)
YearsWithCurrManager	Redundant with YearsAtCompany ($\rho = 0.84$)
Age	Redundant with TotalWorkingYears ($\rho = 0.66$)
PerformanceRating	Only two values (3: 84.6%, 4: 15.4%) and tied to PercentSalaryHike
PercentSalaryHike	Tied to PerformanceRating ($\rho = 0.63$), unrelated to the goal
Education, NumCompaniesWorked, StockOptionLevel, TrainingTimesLastYear	Not linked to the workload, satisfaction or seniority goals
Department, JobRole, EducationField, MaritalStatus, Gender	Nominal; used to profile the clusters after clustering

3.4 Feature Preprocessing for Clustering

3.4.1 Encoding

- **Ordinal encoding.** `BusinessTravel` has a natural order and was mapped to Non-Travel = 0, Travel_Rarely = 1, Travel_Frequently = 2.
- **One-hot encoding.** The five nominal variables (Department, JobRole, EducationField, MaritalStatus, Gender) have no order and were one-hot encoded into 23 binary columns. They are *not* part of the clustering input: they would make up 23 of the 36 columns and dominate the distances, and they lower the silhouette score (Table 15). They are kept to profile the clusters.
- The survey scores and `JobLevel` are already ordered integers and are used as they are.

3.4.2 Log transformation

Continuous variables with skewness above 1 were transformed with $x' = \log(1 + x)$ (Figure 7). The transformation keeps every employee, pulls in the long upper tail, and is safe for the value 0.

Table 14: Skewness before and after the $\log(1 + x)$ transformation

Variable	Before	After
MonthlyIncome	1.37	0.29
TotalWorkingYears	1.12	−0.62
YearsAtCompany	1.76	−0.21
YearsSinceLastPromotion	1.98	0.72

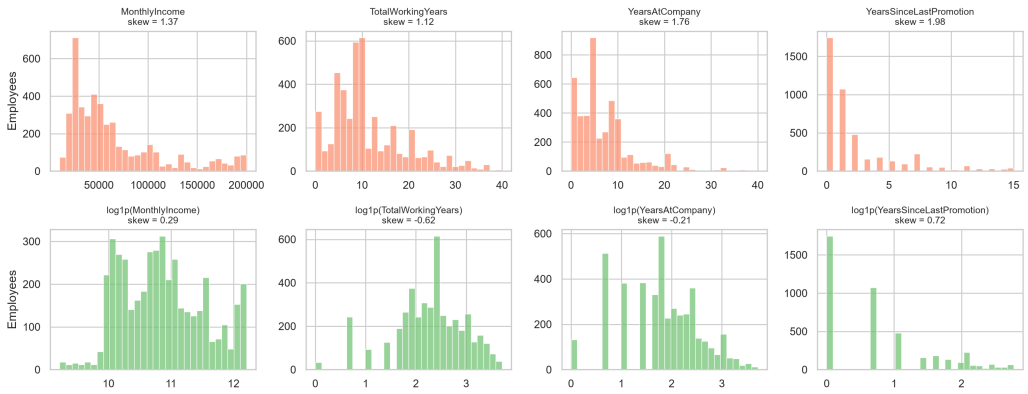


Figure 7: Skewed variables before (top) and after (bottom) the $\log(1+x)$ transformation.

3.4.3 Feature scaling

`StandardScaler` was applied, $z = (x - \mu)/\sigma$, so every feature has mean 0 and variance 1. Without scaling, `MonthlyIncome` (standard deviation 47,069) is thousands of times larger than the other variables (for example 8.11 for `DistanceFromHome`) and decides almost all of the distance. Table 15 confirms the effect: the unscaled data gives a silhouette score of 0.708 for $k = 2$, but this is an artefact of one dominant variable, not a real group structure.

Table 15: K-Means silhouette score by number of clusters k for three preprocessing choices

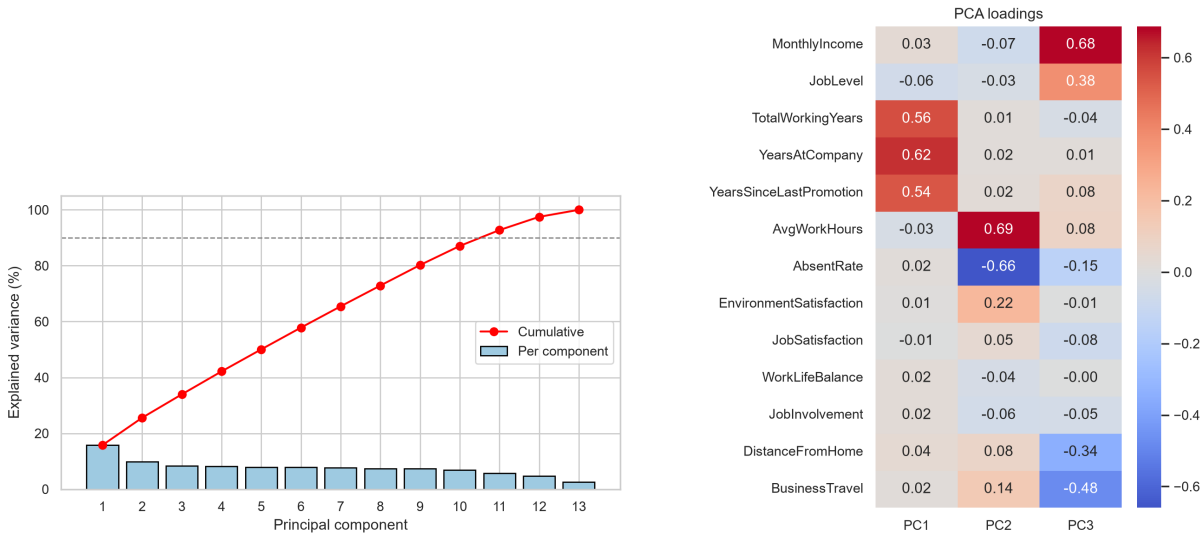
Input	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
Raw features (no scaling)	0.708	0.624	0.610	0.617	0.618
Log + StandardScaler (13 features)	0.103	0.082	0.078	0.082	0.080
Scaled + one-hot nominal (36 columns)	0.083	0.064	0.061	0.064	0.062

`StandardScaler` was preferred over `MinMaxScaler` because the data still contains tail values after the log transformation, and min–max scaling would squeeze most employees into a narrow range.

3.5 Dimensionality Reduction

3.5.1 PCA

PCA was fitted on the 13 scaled features. Figure 8 shows that the first two components explain only 15.8% and 9.8% of the variance (25.6% together), and 11 of the 13 components are needed to reach 90%. The information is therefore spread over many directions, and PCA would remove very little. For this reason PCA is used **for visualisation only**, and the clustering algorithms run on the 13 scaled features.



(a) Explained variance (b) Loadings of PC1–PC3

Figure 8: PCA of the scaled features.

The loadings give each component a clear meaning:

- **PC1 (15.8%) – tenure**: `YearsAtCompany` (0.62), `TotalWorkingYears` (0.56) and `YearsSinceLastPromotion` (0.54).
- **PC2 (9.8%) – workload**: `AvgWorkHours` (0.69) against `AbsentRate` (−0.66).
- **PC3 – pay and travel**: `MonthlyIncome` (0.68), `JobLevel` (0.38), `BusinessTravel` (−0.48) and `DistanceFromHome` (−0.34).

3.5.2 PCA and t-SNE in 2D

Figure 9 projects the employees onto two dimensions with PCA and t-SNE (perplexity 30, `random_state=42`), coloured by `AvgWorkHours`. The colour is only a visual aid and is not used by the models.

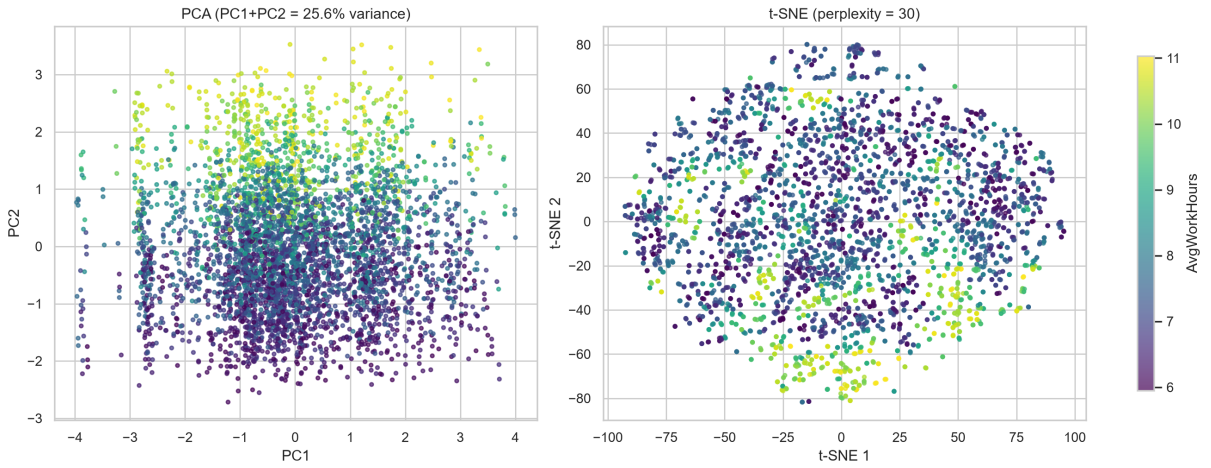


Figure 9: PCA (left) and t-SNE (right) projections of the scaled features, coloured by `AvgWorkHours`.

In the PCA view, working hours increase smoothly along PC2, which matches the loadings above. There is no empty gap between groups: the employees form one continuous cloud. The t-SNE view shows the same picture, with only a few local concentrations of long working hours and no clearly separated islands. t-SNE is used for drawing only; it does not preserve global distances, so it is not used as clustering input.

3.6 Summary of EDA and Feature Selection

- 13 features in four groups (seniority and pay, workload and attendance, satisfaction and engagement, commute and travel) were selected; redundant, constant-like and nominal variables were set aside.
- The input matrix `X` (4,410 × 13) is log-transformed where skewed and standardised; nominal variables are one-hot encoded for profiling only.
- The features are weakly correlated and the 2D projections show one continuous cloud. Clusters with low-to-moderate separation should therefore be expected: the silhouette score of the scaled data is about 0.08–0.10. A low score here reflects the data, not an error, and the clusters will be judged together with the other indices and their business meaning.
- In the next phase K-Means, Hierarchical (Ward) clustering and DBSCAN are run on `X`, and PCA/t-SNE are used to draw the resulting clusters.

4 Clustering Models

Three algorithms were applied to the matrix `X` of 4,410 employees × 13 scaled features (Section 3.4): K-Means, Hierarchical clustering with Ward linkage, and DBSCAN. All

results use the same input and `random_state=42`.

4.1 Evaluation Criteria

Because no label exists, the clusterings are judged with internal indices. For employee i, let $a(i)$ be the mean distance to the other members of its cluster and $b(i)$ the mean distance to the nearest other cluster.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \qquad \text{Silhouette} = \frac{1}{n}\sum_{i=1}^{n} s(i) \in [-1, 1], \tag{4}$$

$$\text{WCSS} = \sum_{j=1}^{k} \sum_{x \in C_j} \|x - \mu_j\|^2 \quad \text{(inertia)}. \tag{5}$$

- **Silhouette** (higher is better): cohesion against separation.
- **Davies–Bouldin** (lower is better): average similarity between each cluster and its most similar cluster.
- **Calinski–Harabasz** (higher is better): ratio of between-cluster to within-cluster dispersion. It tends to fall as k grows, so it is read together with the other indices.
- **Elbow (WCSS)**: the k after which the decrease of inertia slows down.
- **Stability**: mean Adjusted Rand Index (ARI) between K-Means models fitted on 10 random 80% subsamples, each labelling all employees. A value near 1 means the same partition is found again.

4.2 K-Means

K-Means was run for $k = 2, \ldots, 10$ with 20 random initialisations per k. Table 16 and Figure 10 show the indices.

Table 16: K-Means model selection

k	Inertia	Drop (%)	Silhouette	Davies–B.	Calinski–H.	Stability
2	51,465	–	**0.103**	2.716	**502.3**	**0.947**
3	48,751	5.3	0.081	3.046	387.8	0.851
4	46,659	4.3	0.078	2.848	335.9	0.623
5	44,709	4.2	0.082	2.666	310.9	0.816
6	43,145	3.5	0.080	2.570	289.6	0.704
7	41,837	3.0	0.075	2.458	271.8	0.787
8	40,806	2.5	0.078	2.338	254.6	0.683
9	39,982	2.0	0.074	**2.251**	238.7	0.602
10	39,298	1.7	0.066	2.413	224.3	0.562

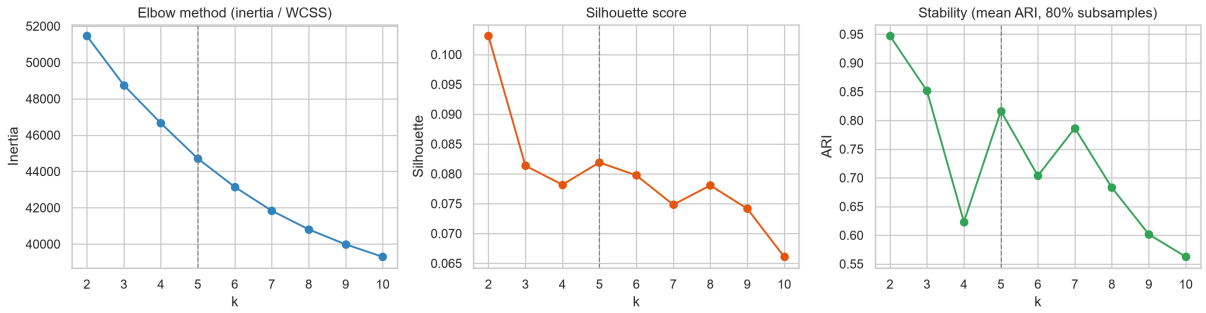


Figure 10: Elbow curve, silhouette score and stability of K-Means for $k = 2, \ldots, 10$. The dashed line marks the chosen $k = 5$.

Reading the indices.

- **Elbow.** The inertia falls smoothly (the drop shrinks from 5.3% to 1.7% per extra cluster) and there is no sharp bend. The elbow method alone therefore does not give a clear k, which is typical of data without well-separated groups.
- **Silhouette.** The global maximum is at $k = 2$ (0.103), followed by a plateau of 0.075–0.082 for $k = 3, \ldots, 8$ with a local peak at $k = 5$ (0.082).
- $k = 2$ **is too coarse.** The two clusters differ almost only in tenure (mean z-scores of `YearsSinceLastPromotion`, `YearsAtCompany` and `TotalWorkingYears` are $-0.53/+1.09$, $-0.48/+0.98$ and $-0.36/+0.74$ in the two clusters, while all other features stay within ± 0.1). This corresponds to the first principal component found in the PCA and cannot represent the workload, commute and satisfaction profiles targeted by the project.
- **Choice.** Restricting the search to $k \geq 3$, $\mathbf{k = 5}$ has the highest silhouette (0.0819 against 0.0814 for $k = 3$), the lowest Davies–Bouldin value among $k = 3, 4, 5$ (2.666), and a high stability (ARI 0.82). $k = 4$ and $k = 6$ are clearly less stable (0.62 and 0.70). The margin between $k = 3$ and $k = 5$ is small; $k = 5$ was kept because it gives finer segments, and $k = 3$ remains a valid coarser alternative (stability 0.85).

Silhouette plot. Figure 11 shows the silhouette of every employee for $k = 5$. The five clusters have sizes 949, 556, 1,469, 740 and 696. The mean silhouette is 0.082 and only 3.2% of employees have a negative value, so few employees are clearly assigned to the wrong cluster; but most values are small (below 0.2), meaning that the clusters touch each other rather than being isolated.

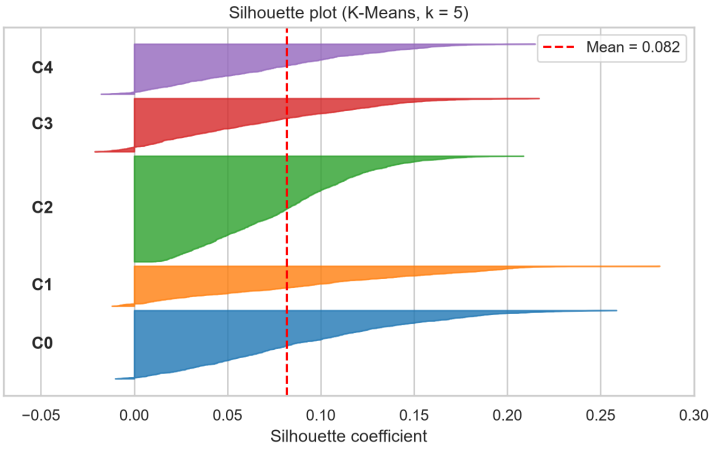


Figure 11: Silhouette plot of K-Means with $k = 5$.

4.3 Hierarchical Clustering (Ward)

Agglomerative clustering with Ward linkage (minimum increase of within-cluster variance, Euclidean distance) was applied to the same matrix. Figure 12 shows the dendrogram of the last 30 merges and the merge distance as a function of the number of clusters.

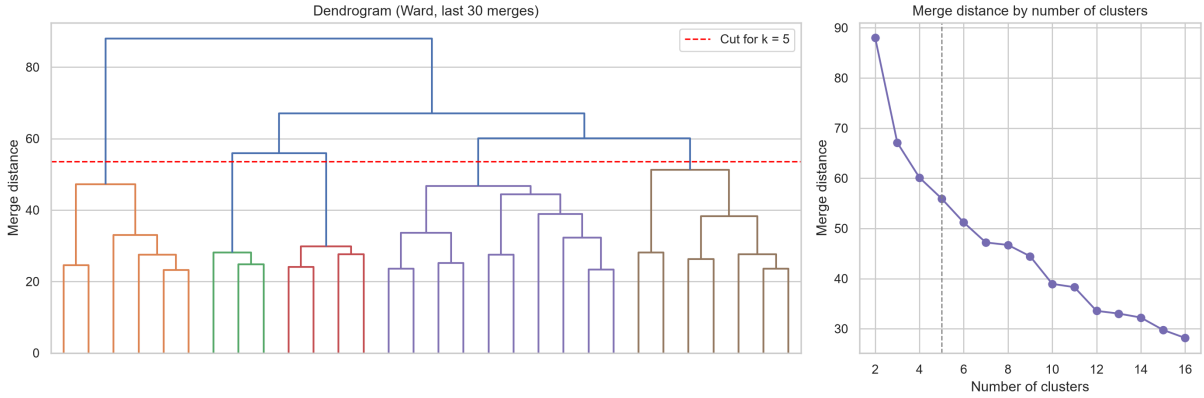


Figure 12: Ward dendrogram (left) and merge distance by number of clusters (right). The dashed line is the cut for $k = 5$.

The merge distance decreases gradually: 88.0 for $k = 2$, 67.1 for $k = 3$, 60.1 for $k = 4$, 56.0 for $k = 5$ and 51.2 for $k = 6$. There is no large jump that would indicate a natural cut, so the dendrogram confirms that the data has a continuous rather than a clearly layered structure. Cutting at five clusters (the choice made for K-Means) gives sizes 989, 451, 434, 1,576 and 960.

Table 17: Ward model selection

k	Silhouette	Davies–B.	Calinski–H.
2	0.073	2.976	319.3
3	0.062	3.378	263.5
4	0.045	3.543	235.8
5	0.047	3.240	218.6
6	0.048	3.060	204.6
7	0.047	2.976	192.6
8	0.044	2.989	184.6
9	0.043	2.861	177.7
10	0.038	2.711	169.5

For every k Ward has a lower silhouette than K-Means (Table 17 against Table 16). The two $k = 5$ partitions agree only partly (ARI $= 0.344$), so Ward does not simply reproduce the K-Means segments.

4.4 DBSCAN

DBSCAN finds dense regions and labels the remaining points as noise. It was used mainly to detect employees with unusual behaviour. With 13 features, `min_samples` was set to $2 \times 13 = 26$. Figure 13 shows the sorted distance of each employee to its 26-th nearest neighbour, which guides the choice of `eps`.

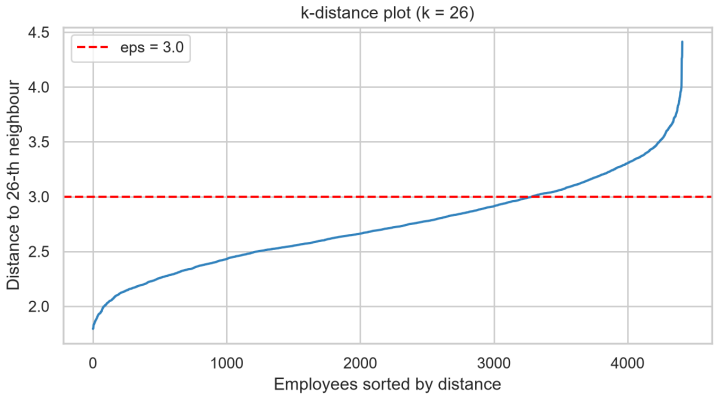


Figure 13: k-distance plot ($k = 26$). The dashed line marks $\varepsilon = 3.0$.

Table 18: DBSCAN sensitivity to ε (`min_samples` $= 26$)

ε	Clusters	Noise points	Noise (%)
2.6	1	927	21.0
2.8	1	348	7.9
3.0	**1**	**105**	**2.4**
3.2	1	33	0.7
3.4	1	4	0.1

For every tested ε DBSCAN returns **one single cluster** plus noise (Table 18). The employees therefore form one connected dense region and no density-separated groups exist, which agrees with the dendrogram and the 2D projections. Because of this, DBSCAN cannot be used to segment the workforce and silhouette-type indices are not defined for it. It is used as an anomaly detector: $\varepsilon = 3.0$ marks 105 employees (2.4%) as noise, a share small enough to be reviewed individually.

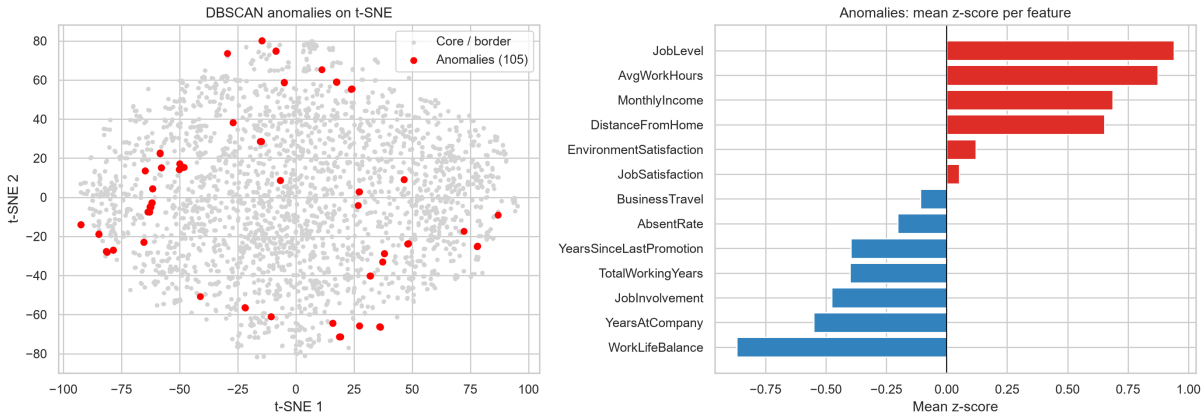


Figure 14: The 105 DBSCAN anomalies on the t-SNE map (left) and their mean z-score per feature (right).

Figure 14 shows what these employees have in common: they are high-level, long-hour employees with poor work-life balance. Compared with the average employee they have a higher `JobLevel` (+0.94), higher `AvgWorkHours` (+0.87), higher `MonthlyIncome` (+0.69) and longer commute (+0.65), and a much lower `WorkLifeBalance` (−0.87), shorter tenure and lower `JobInvolvement` (−0.48). As an external check, their attrition rate is **36.2%**, against 15.6% for the other employees, so the anomalies flag a group at real risk of leaving even though `Attrition` was never used by the algorithm.

4.5 Model Comparison and Selected Model

Table 19: Comparison of the algorithms at $k = 5$

Model	Silhouette	Davies–B.	Calinski–H.	Result
K-Means ($k = 5$)	**0.082**	**2.666**	**310.9**	5 segments
Ward ($k = 5$)	0.047	3.240	218.6	5 segments
DBSCAN ($\varepsilon = 3.0$)	–	–	–	1 cluster + 105 anomalies

K-Means is better than Ward on all three indices, and DBSCAN finds no segmentation. **K-Means with $k = 5$ is therefore selected** to segment the employees, and DBSCAN is kept as a complementary anomaly detector. Figure 15 shows the K-Means clusters on the PCA and t-SNE maps. The clusters are neighbouring regions of one continuous cloud, not separate islands, as expected from the low silhouette.

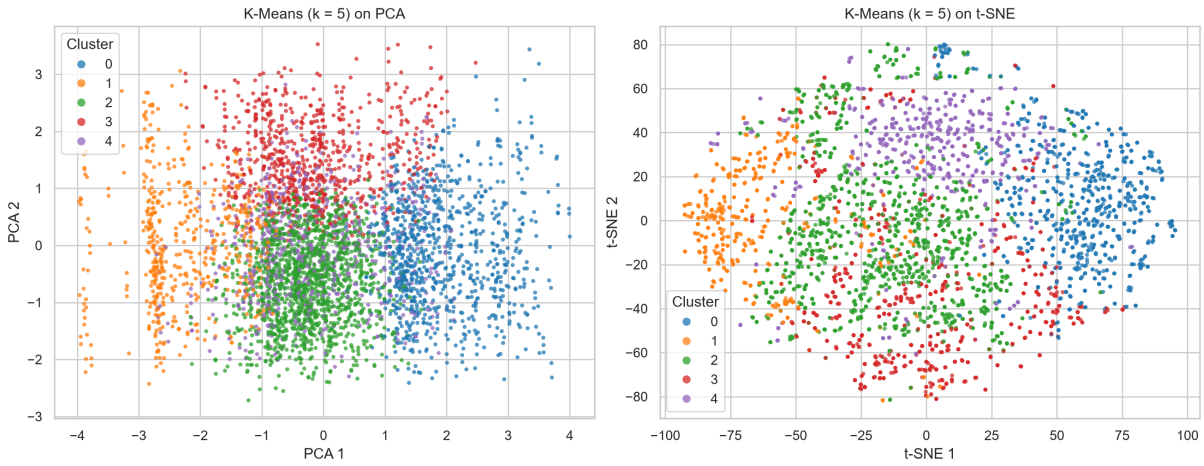


Figure 15: K-Means clusters ($k = 5$) on the PCA (left) and t-SNE (right) maps.

4.6 Summary and Limitations

- K-Means with $k = 5$ is the selected model (silhouette 0.082, Davies–Bouldin 2.666, stability ARI 0.82); $k = 3$ is a close, more stable alternative.
- All indices are low in absolute terms (silhouette below 0.11 for every model and k). The data has weak group structure: the features are almost uncorrelated and the points form one continuous cloud. The clusters should be read as *soft segments* along the main dimensions (tenure, workload, commute), not as natural, isolated groups.
- The indices alone cannot prove that $k = 5$ is the true number of groups; the segments must be judged by their meaning and by external validation, which is done in the next section by profiling them and comparing their attrition rates.
- DBSCAN highlights 105 high-level, long-hour employees with poor work-life balance and a 36.2% attrition rate.

5 Clustering Models

Three algorithms were trained on the matrix `X` of 4,410 employees × 13 scaled features (Section 3.4): K-Means, Hierarchical clustering with Ward linkage, and DBSCAN. All results use the same input and `random_state=42`.

5.1 Evaluation Criteria

Because no label exists, the clusterings are judged with internal indices. For employee i, let $a(i)$ be the mean distance to the other members of its cluster and $b(i)$ the mean distance to the nearest other cluster.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \qquad \text{Silhouette} = \frac{1}{n}\sum_{i=1}^{n} s(i) \in [-1, 1], \tag{6}$$

$$\text{WCSS} = \sum_{j=1}^{k} \sum_{x \in C_j} \|x - \mu_j\|^2 \quad \text{(inertia)}. \tag{7}$$

- **Silhouette** (higher is better): cohesion against separation.
- **Davies–Bouldin** (lower is better): average similarity between each cluster and its most similar cluster.
- **Calinski–Harabasz** (higher is better): ratio of between-cluster to within-cluster dispersion. It tends to fall as k grows, so it is read together with the other indices.
- **Elbow (WCSS)**: the k after which the decrease of inertia slows down.
- **Stability**: mean Adjusted Rand Index (ARI) between K-Means models fitted on 10 random 80% subsamples, each labelling all employees. A value near 1 means the same partition is found again.

5.2 K-Means

K-Means was run for $k = 2, \ldots, 10$ with 20 random initialisations per k. Table 16 and Figure 10 show the indices.

Table 20: K-Means model selection

k	Inertia	Drop (%)	Silhouette	Davies–B.	Calinski–H.	Stability
2	51,465	–	**0.103**	2.716	**502.3**	**0.947**
3	48,751	5.3	0.081	3.046	387.8	0.851
4	46,659	4.3	0.078	2.848	335.9	0.623
5	44,709	4.2	0.082	2.666	310.9	0.816
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7	41,837	3.0	0.075	2.458	271.8	0.787
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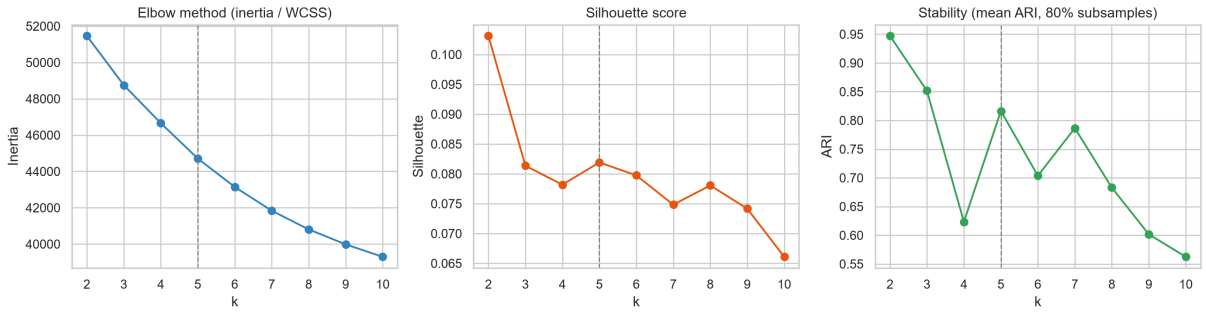


Figure 16: Elbow curve, silhouette score and stability of K-Means for $k = 2, \ldots, 10$. The dashed line marks the chosen $k = 5$.

Reading the indices.

- **Elbow.** The inertia falls smoothly (the drop shrinks from 5.3% to 1.7% per extra cluster) and there is no sharp bend. The elbow method alone therefore does not give a clear k, which is typical of data without well-separated groups.
- **Silhouette.** The global maximum is at $k = 2$ (0.103), followed by a plateau of 0.075–0.082 for $k = 3, \ldots, 8$ with a local peak at $k = 5$ (0.082).
- $k = 2$ **is too coarse.** The two clusters differ almost only in tenure (mean z-scores of `YearsSinceLastPromotion`, `YearsAtCompany` and `TotalWorkingYears` are $-0.53/+1.09$, $-0.48/+0.98$ and $-0.36/+0.74$ in the two clusters, while all other features stay within ± 0.1). This corresponds to the first principal component found in the PCA and cannot represent the workload, commute and satisfaction profiles targeted by the project.
- **Choice.** Restricting the search to $k \geq 3$, $\mathbf{k = 5}$ has the highest silhouette (0.0819 against 0.0814 for $k = 3$), the lowest Davies–Bouldin value among $k = 3, 4, 5$ (2.666), and a high stability (ARI 0.82). $k = 4$ and $k = 6$ are clearly less stable (0.62 and 0.70). The margin between $k = 3$ and $k = 5$ is small; $k = 5$ was kept because it gives finer segments, and $k = 3$ remains a valid coarser alternative (stability 0.85).

Silhouette plot. Figure 11 shows the silhouette of every employee for $k = 5$. The five clusters have sizes 949, 556, 1,469, 740 and 696. The mean silhouette is 0.082 and only 3.2% of employees have a negative value, so few employees are clearly assigned to the wrong cluster; but most values are small (below 0.2), meaning that the clusters touch each other rather than being isolated.

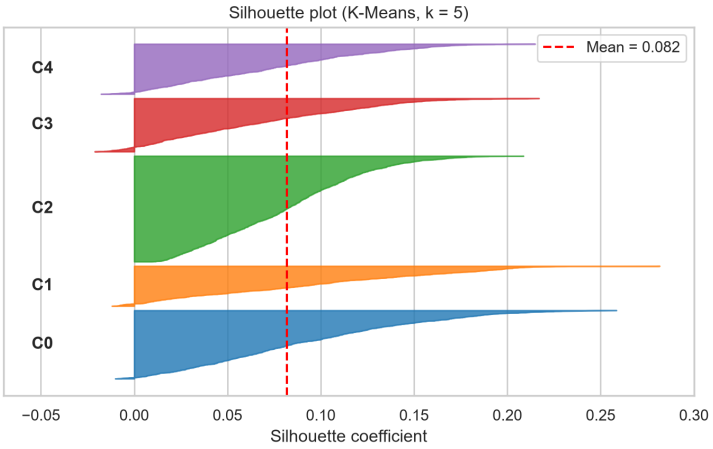


Figure 17: Silhouette plot of K-Means with $k = 5$.

Clusters on the PCA and t-SNE maps. Figure 15 shows the five K-Means clusters on the 2D maps of the previous section. The clusters are neighbouring regions of one continuous cloud, not separate islands, as expected from the low silhouette.

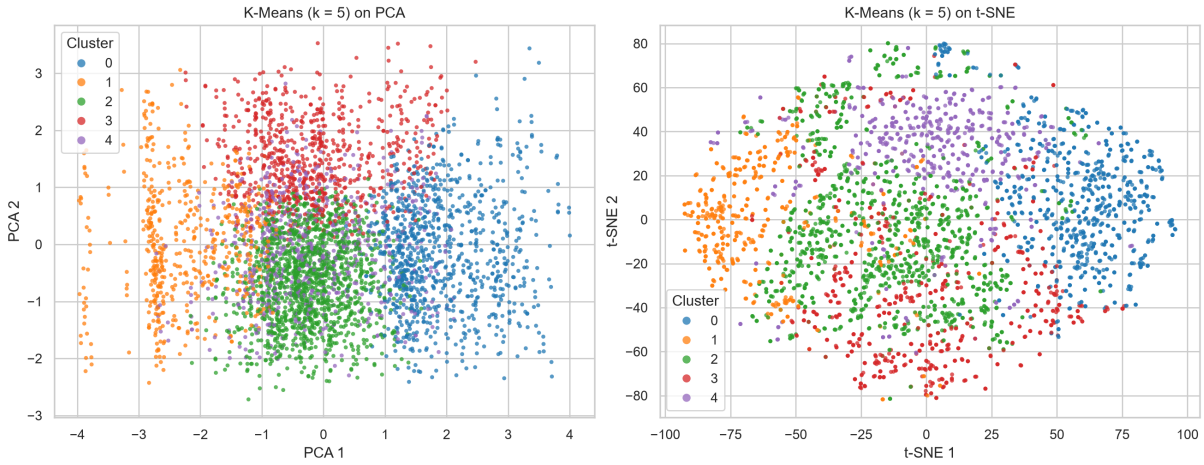


Figure 18: K-Means clusters ($k = 5$) on the PCA (left) and t-SNE (right) maps.

5.3 Hierarchical Clustering (Ward)

Agglomerative clustering with Ward linkage (minimum increase of within-cluster variance, Euclidean distance) was applied to the same matrix. Figure 12 shows the dendrogram of the last 30 merges and the merge distance as a function of the number of clusters.

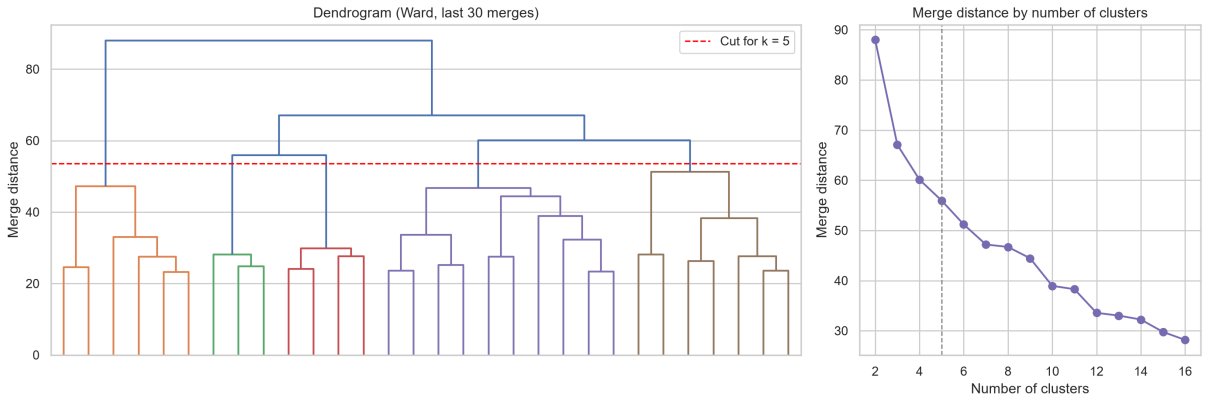


Figure 19: Ward dendrogram (left) and merge distance by number of clusters (right). The dashed line is the cut for $k = 5$.

The merge distance decreases gradually: 88.0 for $k = 2$, 67.1 for $k = 3$, 60.1 for $k = 4$, 56.0 for $k = 5$ and 51.2 for $k = 6$. There is no large jump that would indicate a natural cut, so the dendrogram confirms that the data has a continuous rather than a clearly layered structure. Cutting at five clusters (the choice made for K-Means) gives sizes 989, 451, 434, 1,576 and 960.

Table 21: Ward model selection

k	**Silhouette**	**Davies–B.**	**Calinski–H.**
2	0.073	2.976	319.3
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5.4 DBSCAN

DBSCAN finds dense regions and labels the remaining points as noise. It was used mainly to detect employees with unusual behaviour. With 13 features, `min_samples` was set to $2 \times 13 = 26$. Figure 13 shows the sorted distance of each employee to its 26-th nearest neighbour, which guides the choice of `eps`.

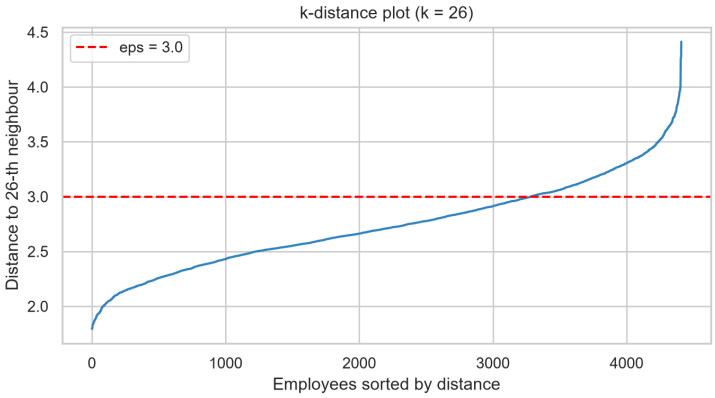


Figure 20: k-distance plot ($k = 26$). The dashed line marks $\varepsilon = 3.0$.

Table 22: DBSCAN sensitivity to ε (`min_samples` = 26)

ε	**Clusters**	**Noise points**	**Noise (%)**
2.6	1	927	21.0
2.8	1	348	7.9
3.0	**1**	**105**	**2.4**
3.2	1	33	0.7
3.4	1	4	0.1

For every tested ε DBSCAN returns **one single cluster** plus noise (Table 18). The employees therefore form one connected dense region and no density-separated groups exist, which agrees with the dendrogram and the 2D projections. Because of this, DBSCAN cannot be used to segment the workforce and silhouette-type indices are not defined for it. It is used as an anomaly detector: $\varepsilon = 3.0$ marks 105 employees (2.4%) as noise, a share small enough to be reviewed individually.

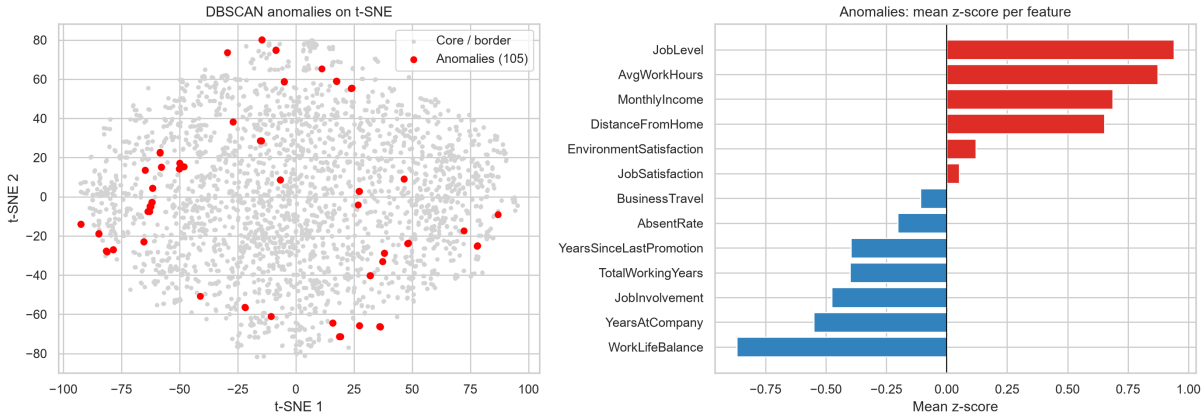


Figure 21: The 105 DBSCAN anomalies on the t-SNE map (left) and their mean z-score per feature (right).

Figure 14 shows what these employees have in common: they are high-level, long-hour employees with poor work-life balance. Compared with the average employee they have a

higher `JobLevel` (+0.94), higher `AvgWorkHours` (+0.87), higher `MonthlyIncome` (+0.69) and longer commute (+0.65), and a much lower `WorkLifeBalance` (−0.87), shorter tenure and lower `JobInvolvement` (−0.48). As an external check, their attrition rate is **36.2%**, against 15.6% for the other employees, so the anomalies flag a group at real risk of leaving even though `Attrition` was never used by the algorithm.

5.5 Summary of Model Training

- **K-Means** was trained for $k = 2, \ldots, 10$; $k = 5$ was chosen (silhouette 0.082, Davies–Bouldin 2.666, stability ARI 0.82), with $k = 3$ as a close and slightly more stable alternative. The five clusters have sizes 949, 556, 1,469, 740 and 696.
- **Ward** clustering shows no natural cut in the dendrogram; the tree was cut at five clusters (sizes 989, 451, 434, 1,576 and 960).
- **DBSCAN** finds a single dense cluster for every tested ε and was used as an anomaly detector: 105 employees (2.4%) at $\varepsilon = 3.0$, mostly high-level, long-hour employees with poor work-life balance and a 36.2% attrition rate.
- All indices are low in absolute terms (silhouette below 0.11 for every k). The data has weak group structure: the features are almost uncorrelated and the points form one continuous cloud. The clusters are therefore read as *soft segments* along the main dimensions (tenure, workload, commute), not as isolated natural groups, and their meaning is examined in the next section.

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